

Assignment 2 Defense Notes - Naive Bayes

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Quick Setup

- Classifier Choice: Naive Bayes
- Live Terminal Output: Yes
- Training File: A2--Bayes-Training set.csv
- Test File: A2--Bayes-TEST set.csv

Who, What, When, Where, Why, How

- Who: DFCA wants a classifier for product order Volume.
- What: Naive Bayes calculates probability scores for each possible Volume class.
- When: Probability tables are learned from the Bayes training set, then applied to the Bayes test set.
- Where: Categorical probabilities, continuous statistics, scores, and predictions are exported to Excel.
- Why: Naive Bayes can handle both categorical attributes and continuous Weight/Material values.
- How: The program multiplies the prior probability, categorical probability components, and continuous probability components for each class.
- Smoothing: Laplace smoothing was used because at least one categorical value/class combination had a zero count.

Formula Checklist

- Entropy/MC = $-\sum(p * \log_2(p))$.
- Gain = $MC(D) - WMC(attribute)$.
- ID3 alpha = $c1 / |F1|$.
- ID3 post-pruning: if $(N - M) / Q \leq gK$, remove subtree.
- Naive Bayes prior = $\text{count(class)} / \text{total records}$.
- Laplace smoothing = $(\text{count} + 1) / (\text{class count} + \text{number of possible values})$.
- Continuous probability = $1/(\sqrt{2*\pi} * \text{std}) * e^{-((\text{value}-\text{mean})^2/(2*\text{std}^2))}$.